

the spread. In this case, unlike in section VII.B.3.a, the rebate and access fee are therefore not neutral. Rather, the floor creates rents that in this case accrue to those liquidity suppliers that are able to get to the front of the queue the fastest. These rents are earned at the expense of liquidity takers and slower liquidity providers.<sup>1023</sup>

For stocks that are not constrained by the tick, rebates and access fees are again on average neutral, as we now show. Consider, for example, a stock with an economic spread of 2 cents. Absent a fee or rebate, the quoted spread would equal the economic spread rounded up to the next smallest tick, or this case, also 2 cents.<sup>1024</sup> Given a 30 mil rebate, a liquidity provider would be willing to quote a spread of 1.4 cents (the economic spread of 2 cents minus twice the 0.3 cents rebate, or 1.4 cents). However, given the tick, it is likely that the quoted spread would be a full 2 cents. If it were any lower, the profit-maximizing liquidity provider would incur a marginal cost (the economic spread of 2 cents) that exceeds the marginal benefit (i.e., the next smaller quoted spread of 1 cent plus twice the 0.3 cent rebate, or 1.6 cents). It is unlikely that the liquidity provider would be willing to do this.<sup>1025</sup> Now, consider the perspective of the market participant taking liquidity. Because the liquidity taker is not typically on both sides of the trade, we use the half-spread to measure their costs due to the bid-ask spread.<sup>1026</sup> This liquidity taker pays the half-spread along with the access fee. In this case, the half-spread is 1 cent and the access fee is 0.3 cents, so the total is 1.3 cents. As stated above, in the absence of rebates, the

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<sup>1023</sup> See Budish Letter and Harris Letter; see also supra notes 999 to 1008, and surrounding discussion.

<sup>1024</sup> This stock may become tick-constrained in the future, and indeed some stocks that have an average spread of 2 cents over a prior period could be tick-constrained during that time.

<sup>1025</sup> Equivalently, the liquidity provider is only willing to quote at 1.4 cents or above, and therefore the quoted spread must be at least 2 cents.

<sup>1026</sup> See section VII.B.3.a.

quoted spread would equal its economic spread of 2 cents. The half-spread would be 1 cent, less than 1.3 cents, meaning that the liquidity taker pays more when there are fees and rebates.

However, consider a stock with an economic spread of 2.5 cents. Given a 30 mil rebate, a liquidity provider would be willing to quote a spread of 1.9 cents (the economic spread of 2.5 cents minus twice the 0.3 cents rebate, or 1.9 cents). Again, given the tick, it is likely that the quoted spread would be a full 2 cents. The liquidity taker would again pay 1.3 cents. In this case, in the absence of rebates, it is likely that the quoted spread would be 3 cents. Otherwise, the profit-maximizing liquidity provider would incur a marginal cost (the economic spread of 2.5 cents) that exceeds the marginal benefit (i.e., the next smallest quoted spread of 2 cents). Given that the quoted spread is 3 cents, the quoted half-spread would be 1.5 cents, so the liquidity taker would likely pay more if there were no fees and rebates.

This same reasoning can be used to show that, except for stocks for which the economic spread is one cent or below, the effect of fees and rebates cancels out mathematically.<sup>1027</sup> The main intuition is that, while liquidity providers will quote one tick lower if the rebate pushes the spread below the next smaller tick, liquidity demanders pay the access fee even if the rebate does not change the quoted spread. We show that the gains to liquidity demanders from this situation are exactly offset by their losses when the rebate does not result in quoting one tick lower, all provided that the economic spread is greater than one cent. Thus, the supply-demand curve

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<sup>1027</sup> Let  $S^*$  equal the economic spread,  $R$  the rebate, and  $t$  the tick size. Then, for any integer  $N \geq 1$ , the liquidity provider collects  $t - 2R$  more in profits under no rebates versus rebates when  $S^* \in (tN, tN + 2R]$ , and  $2R$  less in profits when  $S^* \in (tN + 2R, tN + t]$ . These amounts sum to zero assuming a uniform distribution over the interval. The liquidity taker is on the other side of the trade; in total, the profits to providers are losses to takers, and so these also sum to zero. In contrast, for  $N < 1$ , the liquidity provider collects profits  $2R$  over the interval  $(0, t]$ . The assumption of a uniform distribution over the interval is not necessary for the result. It is a reasonable and standard assumption given the lack of specific information on the properties of  $S^*$  over intervals of length equal to the tick size.

reasoning in section VII.B.3.a is robust to the introduction of the tick, provided that the economic spread is greater than 1 tick. In other words, for stocks for which the economic spread exceeds 1 tick, fees and rebates are neutral on average.

One commenter provides a numerical example that might appear to go against this neutrality result. In particular, the commenter states that, “because rebates also increase depth, it is possible that the costs of access fees for liquidity takers are more than offset by the tighter spreads and depth that they create.”<sup>1028</sup> The numerical example is as follows: A liquidity taker wants to buy 1,000 shares, and the tick size is 1 cent. Without access fees and rebates, the liquidity provider is willing to sell 651 shares at a price of \$10.02 and 349 shares at a price of \$10.03, in which instance the liquidity provider earns \$10.02349 per share.<sup>1029</sup> The liquidity taker then pays an average of \$10.02349 per share to buy the 1,000 shares.<sup>1030</sup> In the presence of an access fee and rebate of 30 mils per share (i.e., \$0.003 per share), the commenter states that the liquidity provider is willing to sell all 1,000 shares at \$10.02, in which instance the liquidity provider earns only \$10.023 per share (i.e., \$10.02 per share + \$0.003 rebate per share = \$10.023 per share). The liquidity taker thus pays only \$10.023 per share to buy the 1,000 shares in the presence of an access fee and rebate of 30 mils (i.e., \$10.02 per share + \$0.003 access fee per share = \$10.023 per share).

However, it is not clear from the example why the liquidity provider would be willing to offer all of the 1,000 shares at \$10.02 per share and earn only \$10.023 per share in the presence

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<sup>1028</sup> See Nasdaq Letter I at 23.

<sup>1029</sup> In the commenter’s example, prices account for the presence of a 1 cent tick. This explains why the liquidity provider is offering shares at \$10.02 and \$10.03 and not at prices in between. The liquidity provider earns:  $(\$10.02 * 651 + \$10.03 * 349) / 1,000 = \$10.02349$  per share to sell 1,000 shares.

<sup>1030</sup>  $(\$10.02 * 651 + \$10.03 * 349) / 1,000 = \$10.02349$  per share to buy 1,000 shares.

of the rebate when it earned \$10.02349 per share absent the rebate. Rather, in the presence of the rebate, the liquidity provider would be willing to sell 951 shares at \$10.02 and 49 shares at \$10.03, in which instance it would also earn \$10.02349 per share on average.<sup>1031</sup> Depth does increase, as the commenter states. However, consistent with the neutrality argument, the liquidity taker pays exactly the same as without the fees and rebates. Thus, while tighter spreads offset the cost of access fees, they do not “more than offset” these fees.<sup>1032</sup>

The previous argument for neutrality pertained to stocks for which the economic spread was greater than one tick. For stocks for which the economic spread is less than 1 tick, however, fees and rebates are not neutral. Rather, because the quoted spread cannot fall to compensate for the rebate, provision of liquidity is overpriced at the spread of one tick. Because the price is artificially high, the supply of liquidity is distorted,<sup>1033</sup> leading to rents for liquidity providers who can get to the top of the queue the fastest. It is harder as a result for slower liquidity providers, such as retail investors and institutions to have their limit orders executed. It is also more expensive for investors seeking to access liquidity. For these stocks, lowering the access fees lowers rents and improves market quality, making it cheaper to transact for investors as a whole.

### C. Baseline

The baseline against which the costs, benefits, and the effects on efficiency, competition, and capital formation of the amendments are measured consists of the current state of the trading

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<sup>1031</sup>  $(951 * \$10.02 + 49 * \$10.03) / 1,000 + \$0.003 = \$10.02349$  per share, where \$0.003 is the rebate per share.

<sup>1032</sup> See section VII.E.3 for discussion of the commenter’s concerns regarding capital formation as it relates to reduced depth at the NBBO.

<sup>1033</sup> Proposing Release, supra note 11, at 80328-29.

environment for NMS stocks, including pricing increments; current practice as it relates to order routing, quotes, fees, and rebates; and availability of data about quotes, fees, and rebates; and the current regulatory framework. The economic analysis appropriately considers existing regulatory requirements, including recently adopted rules, as part of its economic baseline against which the costs and benefits of the amendments are measured.<sup>1034</sup>

Several commenters requested that the Commission consider interactions between the economic effects of the proposed rule and other recent Commission rules.<sup>1035</sup> Since the date of the Proposing Release, the Commission has adopted eight rules mentioned by commenters,<sup>1036</sup> namely the Settlement Cycle Adopting Release,<sup>1037</sup> the February 2024 Form PF Adopting

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<sup>1034</sup> See, e.g., Nasdaq v. SEC, 34 F.4th 1105, 1111–15 (D.C. Cir. 2022). This approach also follows Commission staff guidance on economic analysis for rulemaking. See Current Guidance on Economic Analysis in SEC Rulemaking, (Mar. 16, 2012), available at [https://www.sec.gov/divisions/riskfin/rsfi\\_guidance\\_econ\\_analy\\_secrulemaking.pdf](https://www.sec.gov/divisions/riskfin/rsfi_guidance_econ_analy_secrulemaking.pdf) (“The economic consequences of proposed rules (potential costs and benefits including effects on efficiency, competition, and capital formation) should be measured against a baseline, which is the best assessment of how the world would look in the absence of the proposed action.”); id. at 7 (“The baseline includes both the economic attributes of the relevant market and the existing regulatory structure.”). The best assessment of how the world would look in the absence of the proposed or final action typically does not include recently proposed actions, because that would improperly assume the adoption of those proposed actions.

<sup>1035</sup> See, e.g., CCMR Letter; Independent Trustees Letter; SIFMA Letter I; Citadel Letter I; Equity Market Structure Citadel Letter; ICAN Letter; Virtu Letter II; AIC Letter; AIMA Letter; Antitrust Division of the DOJ Letter; Wagner Letter; Danny Mulson Letter. See also supra notes 135 to 137, and surrounding text, discussing these comments. Many commenters referred to the “Equity Market Structure Proposals” as a group, and we understand that commenters intended this to mean the four rulemaking proposals the Commission issued on Dec. 14, 2022. Of those four, only one was adopted prior to these amendments. One commenter described “Interconnected Rules” to include those four proposed rules issued on Dec. 14, 2023, as well as certain rules that were previously or later adopted, see infra notes 1036 to 1046; and a variety of other proposed rules including Safeguarding Advisory Client Assets, Investment Advisers Act of 1940 Release No. 6384 (Aug. 23, 2023), 88 FR 14,672 (Mar. 9, 2023). See AIC Letter at 1 n.3, 9 n.30 (listing rules and proposed rules, but not explaining a specific connection to the Proposing Release).

<sup>1036</sup> In addition, commenters also mentioned the proposal that was ultimately adopted as Private Fund Advisers; Documentation of Registered Investment Adviser Compliance Reviews, Advisers Act Release No. 6383 (Aug. 23, 2023), 88 FR 63206 (Sept. 14, 2023) (“Private Fund Advisers Adopting Release”). On June 5, 2024, the Fifth Circuit issued a ruling that vacated the rules and amendments adopted in the Private Fund Advisers Adopting Release. Nat’l Ass’n of Priv. Fund Managers v. SEC, 103 F.4th 1097 (2024).

<sup>1037</sup> Shortening the Securities Transaction Settlement Cycle, Securities Exchange Act Release No. 96930 (Feb. 15, 2023), 88 FR 13872 (Mar. 6, 2023) (“Settlement Cycle Adopting Release”). The rules and rule amendments adopted in the Settlement Cycle Adopting Release shorten the standard settlement cycle for

Release,<sup>1038</sup> the May 2023 SEC Form PF Adopting Release,<sup>1039</sup> the Dealer Adopting Release,<sup>1040</sup> the Beneficial Ownership Adopting Release,<sup>1041</sup> Rule 10c-1a Adopting Release,<sup>1042</sup> the Short

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most broker-dealer transactions from two business days after the trade date to one business day after the trade date. To facilitate an orderly transition to a shorter settlement cycle, a new rule also establishes requirements related to completing allocations, confirmations, and affirmations no later than the end of trade date for the processing of institutional transactions subject to the rule; requires registered investment advisers to make and keep records of each confirmation received, and of any allocation and each affirmation sent or received, with a date and time stamp for each allocation and affirmation indicating when it was sent or received; and requires clearing agencies that provide a central matching service to establish, implement, and enforce policies and procedures reasonably designed to facilitate straight-through processing and to file an annual report regarding progress with respect to straight-through processing. With certain exceptions, the rule has a compliance date of May 28, 2024. See Settlement Cycle Adopting Release, section VII.

<sup>1038</sup> Form PF: Reporting Requirements for All Filers and Large Hedge Fund Advisers, Advisers Act Release No. 6546 (Feb. 8, 2024), 89 FR 17984 (Mar. 12, 2024) (“February 2024 Form PF Adopting Release”). The Form PF amendments are designed to enhance the Financial Stability Oversight Council’s ability to monitor systemic risk as well as bolster the SEC’s regulatory oversight of private fund advisers and investor protection efforts. The compliance date for the rule is Mar. 12, 2025. February 2024 Form PF Adopting Release, section II.F.

<sup>1039</sup> Form PF: Event Reporting for Large Hedge Fund Advisers and Private Equity Fund Advisers: Requirements for Large Private Equity Fund Adviser Reporting, Investment Company Act of 1940 Release No. 6297 (May 3, 2023), 88 FR 38146 (June 12, 2023) (“May 2023 SEC Form PF Adopting Release”). The Form PF amendments adopted in May 2023 require large hedge fund advisers and all private equity fund advisers to file reports upon the occurrence of certain reporting events. The compliance dates were Dec. 11, 2023, for the event reports in Form PF sections 5 and 6, and June 11, 2024, for the remainder of the Form PF amendments in the May 2023 SEC Form PF Adopting Release. See May 2023 SEC Form PF Adopting Release, section II.E.

<sup>1040</sup> Further Definition of “As a Part of a Regular Business” in the Definition of Dealer and Government Securities Dealer in Connection with Certain Liquidity Providers, Securities Exchange Act Release No. 34-99477 (Feb. 6, 2024), 89 FR 14938 (Feb. 29, 2024) (“Dealer Adopting Release”). New Rules 3a5-4 and 3a44-2 further define the phrase “as a part of a regular business” as used in the statutory definitions of “dealer” and “government securities dealer.” The compliance date is Apr. 29, 2025, for persons engaging in activities that meet the qualitative factors under the final rules. See Dealer Definition Adopting Release, section II.B.

<sup>1041</sup> Modernization of Beneficial Ownership Reporting, Securities Act of 1933 Release No. 11253 (Oct. 10, 2023), 88 FR 76896 (Nov. 7, 2023) (“Beneficial Ownership Adopting Release”). Among other things, the amendments generally shorten the filing deadlines for initial and amended beneficial ownership reports filed on Schedules 13D and 13G, and require that Schedule 13D and 13G filings be made using a structured, machine-readable data language. The amendments are effective Feb. 5, 2024. The new filing deadline for Schedule 13G will not be required before Sept. 30, 2024, and the rule’s structured data requirements will not be required until Dec. 18, 2024. Beneficial Ownership Adopting Release, section II.G.

<sup>1042</sup> Reporting of Securities Loans, Securities Exchange Act Release No. 98737 (Oct. 13, 2023), 88 FR 75644 (Nov. 3, 2023) (“Rule 10c-1a Adopting Release”). This rule requires any covered person who agrees to a covered securities loan on behalf of itself or another person to report specified information about the covered securities loan to a registered national securities association (currently FINRA is the only registered national securities association)—or rely on a reporting agent to do so—and requires the

Position Reporting Adopting Release,<sup>1043</sup> and the Rule 605 Amendments.<sup>1044</sup> The Commission has also considered the potential effects on entities that are implementing other recently adopted rules during the compliance period for these amendments, including the Treasury Clearing

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registered national securities association to make certain information it receives available to the public. Covered persons will include market intermediaries, securities lenders, and broker-dealers, while reporting agents include certain brokers, dealers, or registered clearing agencies. The rule's compliance dates require that the registered national securities association propose rules pursuant to Rule 10c-1a(f) by May 2, 2024, and the proposed rules shall be effective no later than Jan. 2, 2025; that covered persons report Rule 10c-1a information to a registered national securities association on or by Jan. 2, 2026 (which requires that the registered national securities association have implemented data retention and availability requirements for reporting); and that the registered national securities association publicly report Rule 10c-1a information by Apr. 2, 2026. Rule 10c-1a Adopting Release, section VIII.

<sup>1043</sup> Short Position and Short Activity Reporting by Institutional Investment Managers, Securities Exchange Act Release No. 98738 (Oct. 13, 2023), 88 FR 75100 (Nov. 1, 2023) ("Short Position Reporting Adopting Release"). Under the new rule, institutional investment managers that meet or exceed certain specified reporting thresholds are required to report, on a monthly basis using the related form, specified short position data and short activity data for equity securities. The compliance date is Jan. 2, 2025. See Short Position Reporting Adopting Release, section VI. In addition, the Commission adopted an amendment to the national market system ("NMS") plan governing the consolidated audit trail ("CAT") created pursuant to the Exchange Act to require the reporting of reliance on the bona fide market making exception in the Commission's short sale rules. The Commission published the text of the amendment to the NMS plan governing the CAT ("CAT NMS Plan") in a separate notice. The compliance date for the amendment to the CAT NMS Plan is July 1, 2025. See SEC, Notice of the Text of the Amendment to the National Market System Plan Governing the Consolidated Audit Trail for Purposes of Short Sale-Related Data Collection, Securities Exchange Act Release No. 98739 (Oct. 13, 2023), 88 FR 75079 (Nov. 1, 2023).

<sup>1044</sup> Rule 605 Amendments, supra note 10. The Commission adopted amendments to rules requiring disclosures for order executions in NMS stocks, including expanding the scope of reporting entities, modifying the scope of orders covered by the rule, and modifying the information required to be reported under the rule. The rule has an effective date of June 14, 2024, and, with a few exceptions, a compliance date of Dec. 14, 2025. See Rule 605 Amendments, section VII.

Adopting Release<sup>1045</sup> and the Customer Notification Adopting Release.<sup>1046</sup> These recently adopted rules were not included as part of the baseline in the Proposing Release because they were not yet adopted at that time, but they are part of the baseline in this analysis.<sup>1047</sup> In response to commenters, this economic analysis considers potential economic effects arising from any

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<sup>1045</sup> Standards for Covered Clearing Agencies for U.S. Treasury Securities and Application of the Broker-Dealer Customer Protection Rule with Respect to U.S. Treasury Securities, Securities Exchange Act Release No. 99149 (Dec. 13, 2023), 89 FR 2714 (Jan. 16, 2024) (“Treasury Clearing Adopting Release”). Among other things, the amendments require covered clearing agencies for U.S. Treasury securities to have written policies and procedures reasonably designed to require that every direct participant of the covered clearing agency submit for clearance and settlement all eligible secondary market transactions in U.S. Treasury securities to which it is a counterparty. The compliance date was Mar. 18, 2024, for covered clearing agencies to file any proposed rule changes pursuant to Rules 17Ad-22(e)(6)(i), 17Ad-22(e)(18)(iv)(C), and 15c3-3, which must be effective by Mar. 31, 2025. With respect to the changes to Rule 17Ad-22(e)(18)(iv)(A) and (B), (i) covered clearing agencies were required to file any proposed rule changes regarding those amendments no later than June 14, 2024, and (ii) those changes must be effective by Dec. 31, 2025, for cash market transactions encompassed by section (ii) of the definition of an eligible secondary market transaction, and by June 30, 2026, for repo transactions encompassed by section (i) of the definition of eligible secondary market transactions. Finally, the Commission amended the broker-dealer customer protection rule to permit margin required and on deposit with covered clearing agencies for U.S. Treasury securities to be included as a debit in the reserve formulas for accounts of customers and proprietary accounts of broker-dealers, subject to certain conditions. Compliance by the direct participants of a U.S. Treasury securities covered clearing agency with the requirement to clear eligible secondary market transactions is not required until Dec. 31, 2025, and June 30, 2026, respectively, for cash and repo transactions. See Treasury Clearing Adopting Release, section III.

<sup>1046</sup> Regulation S-P: Privacy of Consumer Financial Information and Safeguarding Customer Information, Securities Exchange Act Release No. 100155 (May 15, 2024), 89 FR 47688 (June 3, 2024) (“Customer Notification Adopting Release”). The Commission amended Regulation S-P to require brokers, dealers, investment companies, registered investment advisers, and transfer agents registered with the Commission or another appropriate regulatory agency to adopt written policies and procedures for incident response programs to address unauthorized access to or use of customer information. These must include procedures for providing timely notification to individuals affected by an incident involving sensitive customer information with details about the incident and information designed to help affected individuals respond appropriately. Among other things, the amendments also broadened the scope of information covered by the safeguards rule and the disposal rule, and extended the requirements to safeguard customer records and information to all transfer agents. The compliance date for larger entities is Dec. 3, 2025, and for smaller entities, June 3, 2026. Customer Notification Adopting Release, section II.F.

<sup>1047</sup> Some commenters assumed that the four proposed rules issued on Dec. 14, 2022, would be implemented simultaneously, and therefore stated that the baseline in the Proposing Release was inaccurate to the extent it did not contemplate that the other rules have gone into effect. See, e.g., Equity Market Structure Citadel Letter at 15. As discussed above, however, our baseline does not assume the adoption of proposed rules. Instead, the baseline changes incrementally with each adopted rule. To the extent those or other proposals are adopted in the future, the baseline in those subsequent rulemakings will reflect the existing regulatory requirements at that time.



overlap in compliance dates between these amendments and the other recent amendments.<sup>1048</sup> It also considers interactions between these amendments and the Rule 605 Amendments.<sup>1049</sup>

## **1. Tick Sizes**

Preexisting Rule 612 of Regulation NMS restricts the ability of venues to display, rank, or accept quotations in NMS stocks beyond a certain minimum quoting increment (or tick). This section discusses current regulation on tick sizes and Commission analysis showing that the current tick size acts as a binding price floor on the quoted spread a significant portion of the time for a large fraction of the share volume transacted in NMS stocks.

### **a. Current Regulations**

Preexisting Rule 612 of Regulation NMS, which came into effect on August 29, 2005, prohibited a national securities exchange, national securities association, ATS, vendor, or broker or dealer from displaying, ranking, or accepting quotations, orders, or indications of interest in any NMS stock priced in an increment smaller than \$0.01 if the quotation, order, or indication of interest is priced equal to or greater than \$1.00 per share. If the quotation, order, or indication of interest is priced less than \$1.00 per share, the minimum pricing increment is \$0.0001. Most listing exchanges require stocks listed on their exchanges to maintain a price greater than \$1.00 per share, and consequently \$0.01 is the prevailing tick size for most quotes and orders for NMS

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<sup>1048</sup> See infra section VII.D.6.b.

<sup>1049</sup> See, e.g., SIFMA Letter II (stating that variable tick sizes could diminish the ability to compare execution quality using the Rule 605 disclosures); Citadel Letter II (stating execution quality statistics are important to understanding the effects of these amendments on market quality); AIMA Letter (suggesting finalization of the proposed Rule 605 amendments would, along with the MDI round lot order definition, provide a much more informed economic baseline against which to assess other equity market structure proposals); see also infra section VII.D.6.a, discussing this topic.

stocks.<sup>1050</sup> Preexisting Rule 612 of Regulation NMS effectively establishes \$0.01 as the minimum spread that can be quoted for stocks priced equal to, or greater than, \$1.00 per share because the NBBO is determined by the best displayed round lot quotes, and exchanges are required to have rules in place to avoid and reconcile locked and crossed quotations.<sup>1051</sup>

While preexisting Rule 612 of Regulation NMS restricts quoting or submitting orders in sub-penny increments for NMS stocks priced greater than or equal to \$1.00, it does not restrict trading in sub-penny increments. Sub-penny trading on exchanges and ATSS occurs primarily as a result of midpoint orders and benchmark trades. Benchmark trades, such as volume weighted average price (“VWAP”) and time weighted average price (“TWAP”) orders, may not be explicitly priced in an impermissible sub-penny increment, but the ultimately determined execution price may be in a sub-penny increment. Trading at sub-penny increments also occurs as a result of broker-dealers, including some OTC market makers known as wholesalers, internalizing customer order flow at sub-penny prices.<sup>1052</sup>

Sub-penny trading on registered exchanges may also occur as a result of their RLPs. The Commission granted exemptions from Rule 612 to various national securities exchanges’ RLPs as a means to allow them to compete with OTC sub-penny price improvement.<sup>1053</sup> Under the

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<sup>1050</sup> See, e.g., NYSE Continued Listing Standards, § 802.01C, available at <https://www.nyse.com/listings/resources>; Rulebook - The Nasdaq Stock Market, §5400, available at <https://listingcenter.nasdaq.com/rulebook/nasdaq/rules>.

<sup>1051</sup> See Reg NMS Rule 610(d). A locked market occurs when the bid and ask price for a security are identical. A crossed market occurs when the bid is higher than the ask.

<sup>1052</sup> The term “wholesaler” is not defined in Regulation NMS, but commonly refers to a broker-dealer acting as an OTC market maker that primarily focuses on attracting orders from broker-dealers that service the accounts of a large number of individual investors, referred to in this release as “retail brokers.”

<sup>1053</sup> See Securities Exchange Act Release No. 67347 (July 3, 2012), 77 FR 40673 (July 10, 2012) (approving retail liquidity programs on a pilot basis for NYSE and NYSE Amex and granting rule 612 exemption) (“NYSE Retail Liquidity Program Approval Order”); see also CBOE BYX Rule 11.24; Securities Exchange Act Release No. 68303 (Nov. 27, 2012), 77 FR 71652 (Dec. 3, 2012) (CBOE BYX Retail Pilot